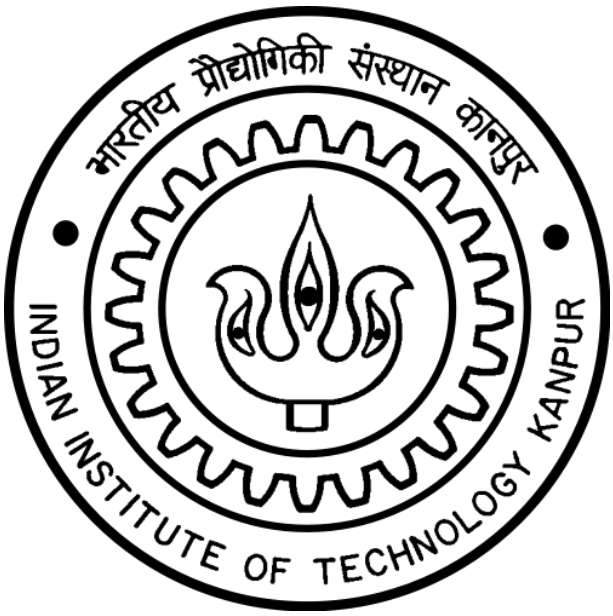


SPA617

RADIO ASTRONOMY LABORATORY

ANTENNA CHARACTERIZATION

TANISHQUE KUMAR KANWAR

M.TECH, SPASE

261310019

1 Antenna Characterization

1.1 Aim

To characterize the given antenna by measuring its beam pattern in both co-polarization and cross-polarization configurations.

1.2 Objectives

(a) To determine the antenna operating frequency and return loss from VSWR measurements.

(b) To measure the effect of distance on the power received by the antenna.

(c) To measure the far-field power pattern of the given antenna with respect to angle in both co-polarization and cross-polarization configurations.

(d) To determine the half-power beamwidth and estimate the directivity of the antenna.

1.3 Theory

1.3.1 Basic antenna concepts

An antenna is a transducer that converts electrical power into electromagnetic waves and vice versa. An antenna can be used either as a transmitting antenna or as a receiving antenna.

A dipole antenna has two equal metal arms in a straight line and is fed at the centre. Its total length is approximately half a wavelength ($\lambda/2$), with a feed line connected between the two arms, as shown in Fig. 1. It radiates most strongly in directions perpendicular to the wire and has very little radiation along the wire.

A monopole antenna is approximately half of a dipole mounted above a conducting ground plane. Its length is approximately one-quarter wavelength ($\lambda/4$). The ground acts like a mirror, making it behave like a full dipole.

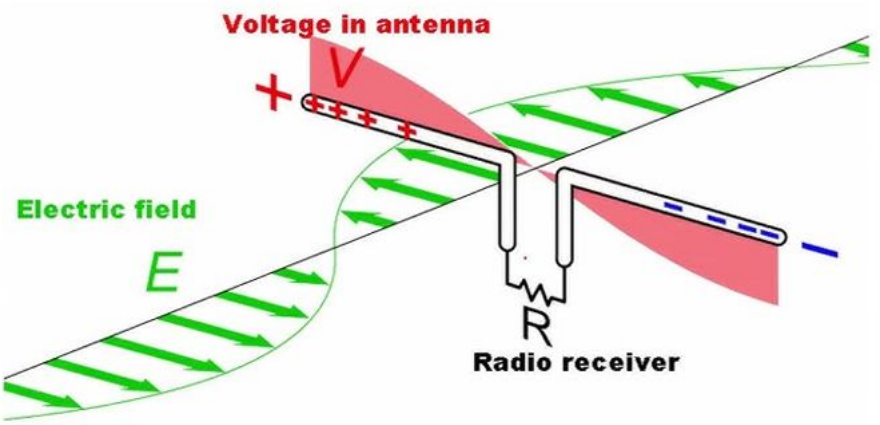


Figure 1: Half-wave dipole antenna as a receiver (image credit: Wikipedia).

- A transmitting antenna converts electrical signals into electromagnetic waves and radiates them.
- A receiving antenna converts the received electromagnetic waves into electrical signals.

1.3.2 Antenna beam pattern

The energy radiated by an antenna is represented by the radiation pattern of the antenna. The radiation pattern generally has a main lobe, side lobes, and a back lobe.

- The **main lobe** is the region where the maximum radiated energy exists. The direction of this lobe indicates the principal direction of the antenna beam.
- The other parts of the pattern where radiation is distributed sideways are known as **side lobes** or minor lobes.
- The lobe opposite to the main lobe is known as the **back lobe**.

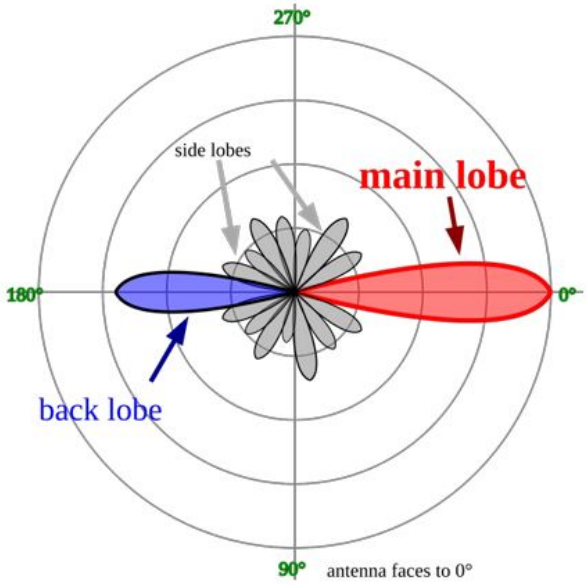


Figure 2: Antenna lobes (image credit: Wikipedia).

E-plane and H-plane angular measurements are used to determine the radiation beam pattern of an antenna. The E-plane is the plane containing the electric-field vector and the propagation vector, while the H-plane is the plane containing the magnetic-field vector and the propagation vector. The E-plane and H-plane are perpendicular to each other.

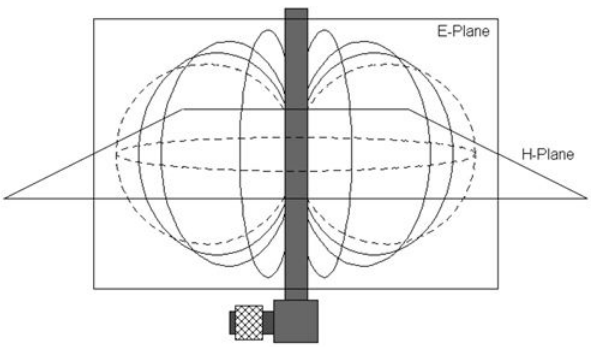


Figure 3: E-plane and H-plane for a vertically polarized monopole antenna (image credit: Wikipedia).

1.3.3 Reflection coefficient, VSWR, and return loss

When the antenna impedance Z_L differs from the characteristic impedance Z_0 of the transmission line, a part of the incident signal is reflected. The voltage reflection coefficient at the antenna terminals is

$$\Gamma = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{Z_L - Z_0}{Z_L + Z_0}. \tag{1}$$

The magnitude $|\Gamma|$ indicates the degree of impedance mismatch. For a perfectly matched antenna,

$$Z_L = Z_0 \implies |\Gamma| = 0. \tag{2}$$

The voltage standing-wave ratio is related to the magnitude of the reflection coefficient by

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{3}$$

The return loss is defined as

$$\text{RL} = -20 \log_{10} |\Gamma| \quad \text{dB}. \tag{4}$$

A well-matched antenna has a VSWR close to unity, a small reflection coefficient, and a large return-loss value. The resonant or operating frequency is generally identified near the minimum of the VSWR curve.

1.3.4 Half-power beamwidth

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two points where the received power drops to half of its maximum value, corresponding to the -3 dB points.

- A small HPBW indicates a narrow, focused beam.
- A large HPBW indicates a broad beam.

1.3.5 Directivity

Directivity is a measure of how concentrated the radiation is in the direction of the main lobe compared with an isotropic antenna. The directivity is related to the beam solid angle Ω_A as

$$D = \frac{4\pi}{\Omega_A}. \tag{5}$$

For practical antennas, the beam solid angle can be approximated using the half-power beamwidths in the E-plane (θ_E) and H-plane (θ_H):

$$D \approx \frac{4\pi}{\theta_E \, \theta_H} \tag{6}$$

where θ_E and θ_H are expressed in radians.

1.4 Components Required

- Monopole and dipole antennas
- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser

- RF Tee connector
- $50\,\Omega$ termination
- Antenna support stand
- Coaxial cables and suitable RF adapters

1.5 Instrument Settings and Precautions

- Keep the signal-generator amplitude constant during each frequency sweep.
- Set the oscilloscope input impedance to $1\,\mathrm{M}\Omega$ for the Tee-junction measurement.
- Use an input signal level that is clearly measurable but does not overdrive the oscilloscope or spectrum analyser.
- Maintain the same antenna height and alignment during each set of measurements.
- Keep the transmitting frequency and power fixed during distance and angular measurements.
- Keep nearby reflecting objects away from the antenna path as far as practicable.

1.6 Experimental Setup

Connect the transmitting antenna to the signal generator and the receiving antenna to the spectrum analyser, as shown in Fig. 6. Keep the antennas at the same height and align them appropriately for the measurement being performed.

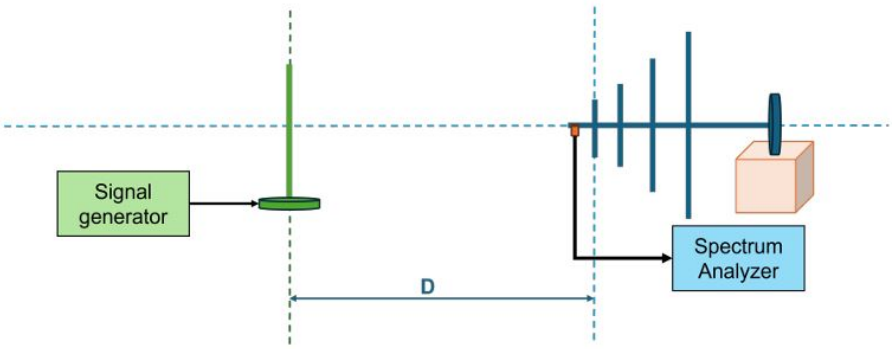


Figure 4: Antenna characterization experimental setup.

For the VSWR and return-loss measurement, connect the signal generator, RF Tee connector, antenna (DUT), and oscilloscope as shown in Fig. **??**. The oscilloscope measures the variation produced by the incident and reflected signals at the Tee junction.

1.7 Experimental Procedures

1.7.1 Part A: VSWR and antenna operating frequency

1. Connect the signal generator, RF Tee connector, antenna (DUT), and oscilloscope as given in Fig. **??**.
2. Set the oscilloscope input impedance to $1\,\mathrm{M}\Omega$.

3. Keep the signal-generator amplitude constant and select test frequencies across the expected operating range of the antenna.

4. Around each selected test frequency f_k, perform a small-span frequency sweep containing one complete amplitude-ripple period.

5. During the sweep, record:

   - V_{max} the maximum voltage amplitude observed on the oscilloscope;
   - V_{min} the minimum voltage amplitude observed on the oscilloscope.

   Use the same voltage quantity for both measurements. For example, if V_{max} is measured as a peak-to-peak voltage, measure V_{min} as a peak-to-peak voltage also.

6. Calculate the measured voltage standing-wave ratio:

$$\text{VSWR} = \frac{V_{\text{max}}}{V_{\text{min}}} \tag{7}$$

7. Calculate the magnitude of the reflection coefficient:

$$|\Gamma| = \frac{\text{VSWR} - 1}{\text{VSWR} + 1} = \frac{V_{\text{max}} - V_{\text{min}}}{V_{\text{max}} + V_{\text{min}}} \tag{8}$$

8. Calculate the return loss:

$$\text{RL} = -20 \log_{10} |\Gamma| \quad \text{dB}. \tag{9}$$

9. Repeat the measurement at different frequencies across the expected operating range of the antenna.

10. Plot VSWR and return loss versus frequency.

11. Identify the frequency at which the VSWR is minimum and state it as the approximate antenna operating or resonant frequency.

12. Determine the frequency range over which VSWR ≤ 2.

Table 1: VSWR and return loss measurement across frequency.

Frequency (MHz)	V_{max} **(V)**	V_{min} **(V)**	**VSWR**	**Return Loss (dB)**
40	485	340	1.42	-15.21
60	485	377	1.28	-18.21
80	456	377	1.20	-20.82
110	456	246	1.85	-10.50
130	327	246	1.32	-17.20
150	327	240	1.36	-16.33
170	304	240	1.26	-18.78
190	304	190	1.6	-12.73
220	207	190	1.08	-28.29
250	207	161	1.28	-18.21

VSWR is less than 2 over the frequency range from 40 MHz to 250 MHz.

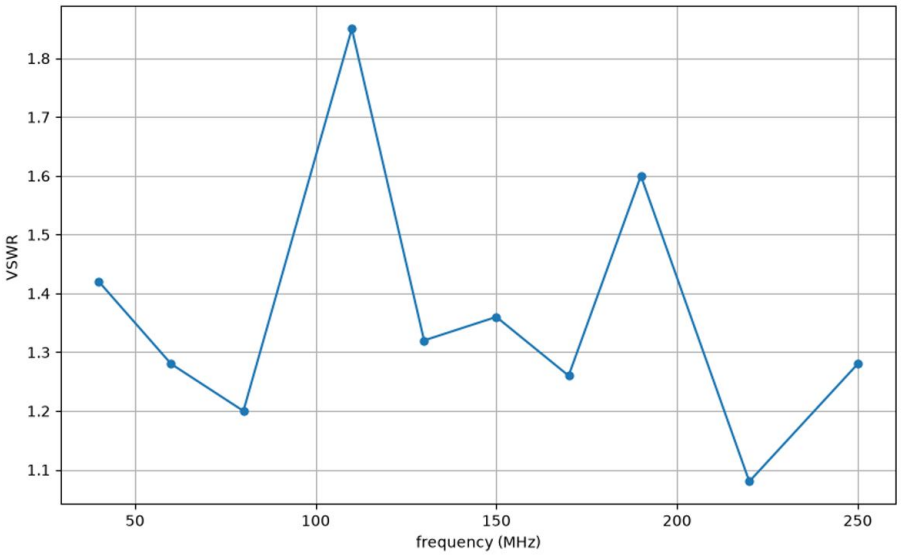


Figure 5: Plot of Frequency (MHz) vs VSWR

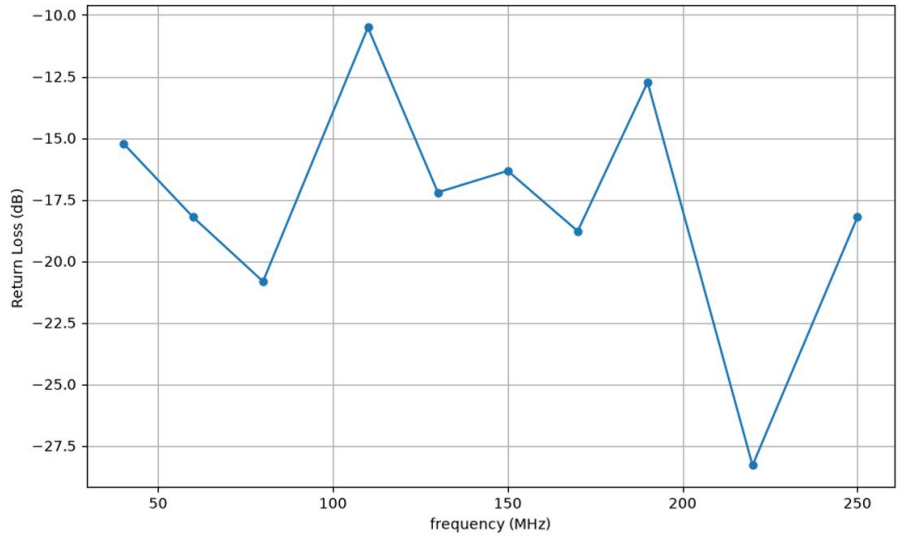


Figure 6: Plot of Frequency (MHz) vs Return Loss (dB)

Compare the measured operating frequency with the theoretical resonance frequency of a $\lambda/4$ monopole or a $\lambda/2$ dipole, as applicable.

λ/4 = 0.05cm = 0.2m

$f_{operating} = \frac{c}{\lambda} = 1.5GHz$

To ensure proper beam pattern measurements, the minimum far-field distance was verified using the maximum aperture dimension (D=0.29m) and the signal generator power set to 20dBm :

$R_{far-field} = 0.841m$

Thus, all angular measurements are taken beyond this 84.1cm boundary.

1.7.2 Part B: Effect of distance

1. Connect the antennas as shown in Fig. 6.
2. Set a suitable frequency and amplitude in the signal generator within the operating range of the antenna.
3. Adjust the length of the monopole according to the selected frequency.
4. Keep the transmitting and receiving antennas aligned with the same polarization and at the same height.
5. Vary the distance between the transmitting and receiving antennas and record the received power.
6. Plot the received power P_r versus distance R.
7. Plot P_r versus $20 \log_{10} R$ and comment on the observed trend.

Table 2: Received power at varying distances between transmitting and receiving antenna

Distance (R) **[cm]**	$20 \log_{10} R$	**Received Power** (P_r) **[dBm]**
118	41.44	-19.38
221	46.89	-26.38
318	50.05	-27.80
416	52.38	-28.27
515	54.24	-29.91
615	55.78	-34.08
715	57.09	-39.03

The received power was plotted against $20log_{10}R$. The resulting linear relationship demonstrates that the path loss of the signal mathematically follows the **inverse-square law** as the distance between the antennas increases.

1.7.3 Part C: Beam pattern – co-polarization measurement

Keep the distance between the transmitting and receiving antennas at a minimum of 3λ. Align the electric fields of the transmitting and receiving antennas in the same plane and keep the antennas at the same height, as shown in Fig. 8.

1. Set a fixed transmitting frequency within the operating range of the antenna.

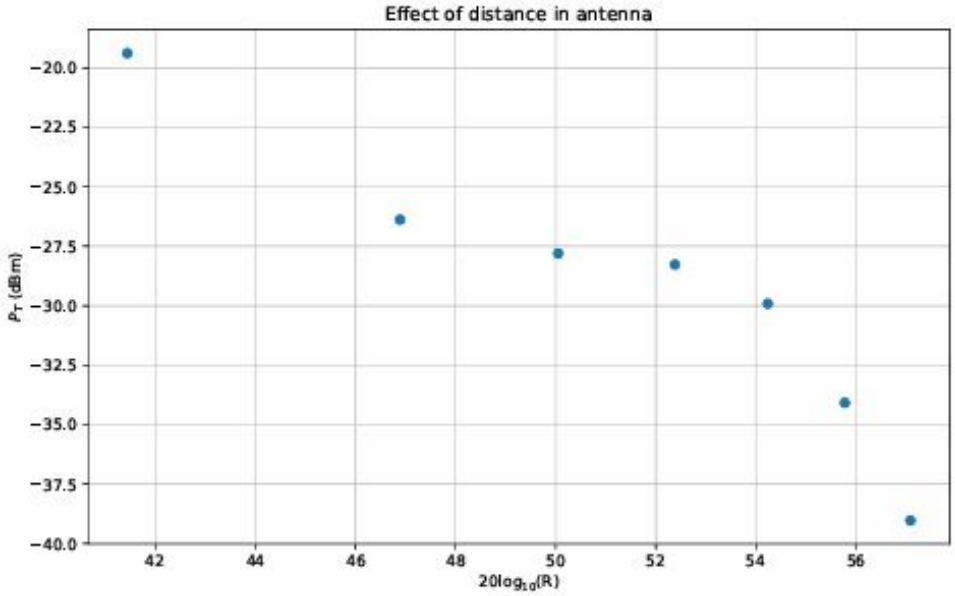


Figure 7: P_r vs $20 \log_{10} R$

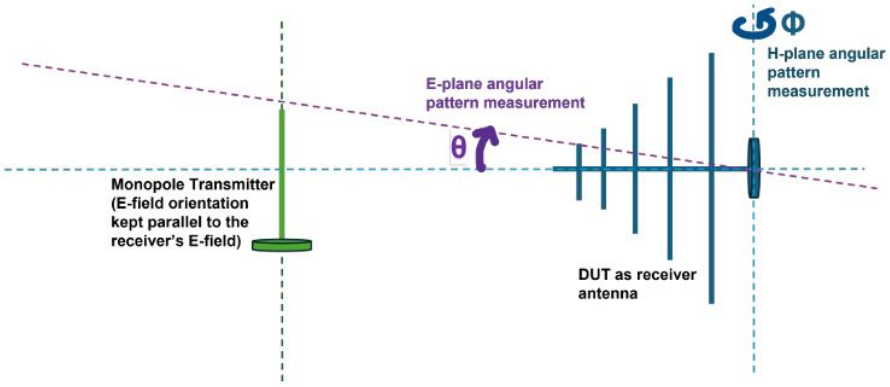


Figure 8: Beam-pattern measurement setup for co-polarization.

2. Vary the E-plane angle from -90° to $+90^\circ$ and record the received power at suitable angular intervals.
3. Return the antenna to the reference position.
4. Vary the H-plane angle from -90° to $+90^\circ$ and record the received power.
5. Plot the E-plane and H-plane beam patterns as polar plots.
6. Repeat the experiment for two additional frequency values and compare the plots.

1.7.4 Part D: Beam pattern – cross-polarization measurement

Adjust the transmitting antenna so that the electric fields of the transmitting and receiving antennas are perpendicular, as shown in Fig. 9.

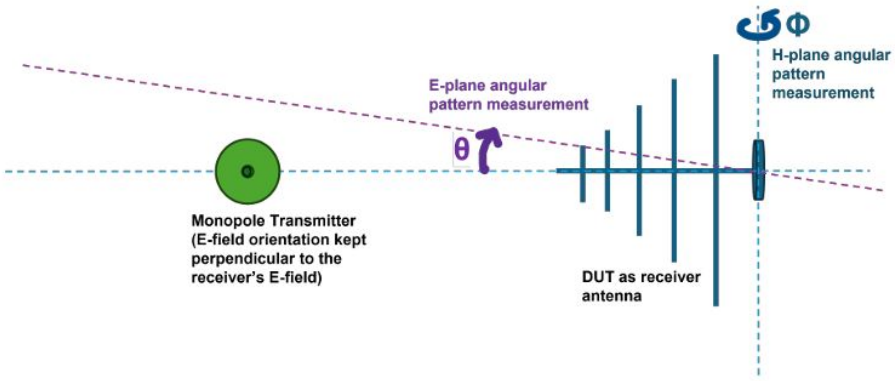


Figure 9: Beam-pattern measurement setup for cross-polarization.

1. Retain the same frequency, antenna separation, heights, angular steps, and instrument settings used for the co-polarization measurement.
2. Vary the E-plane angle and record the received power.
3. Vary the H-plane angle and record the received power.
4. Plot the E-plane and H-plane beam patterns for the cross-polarization configuration.

The radiation patterns were visualized using polar projections to compare the spatial distribution of the main and minor lobes.

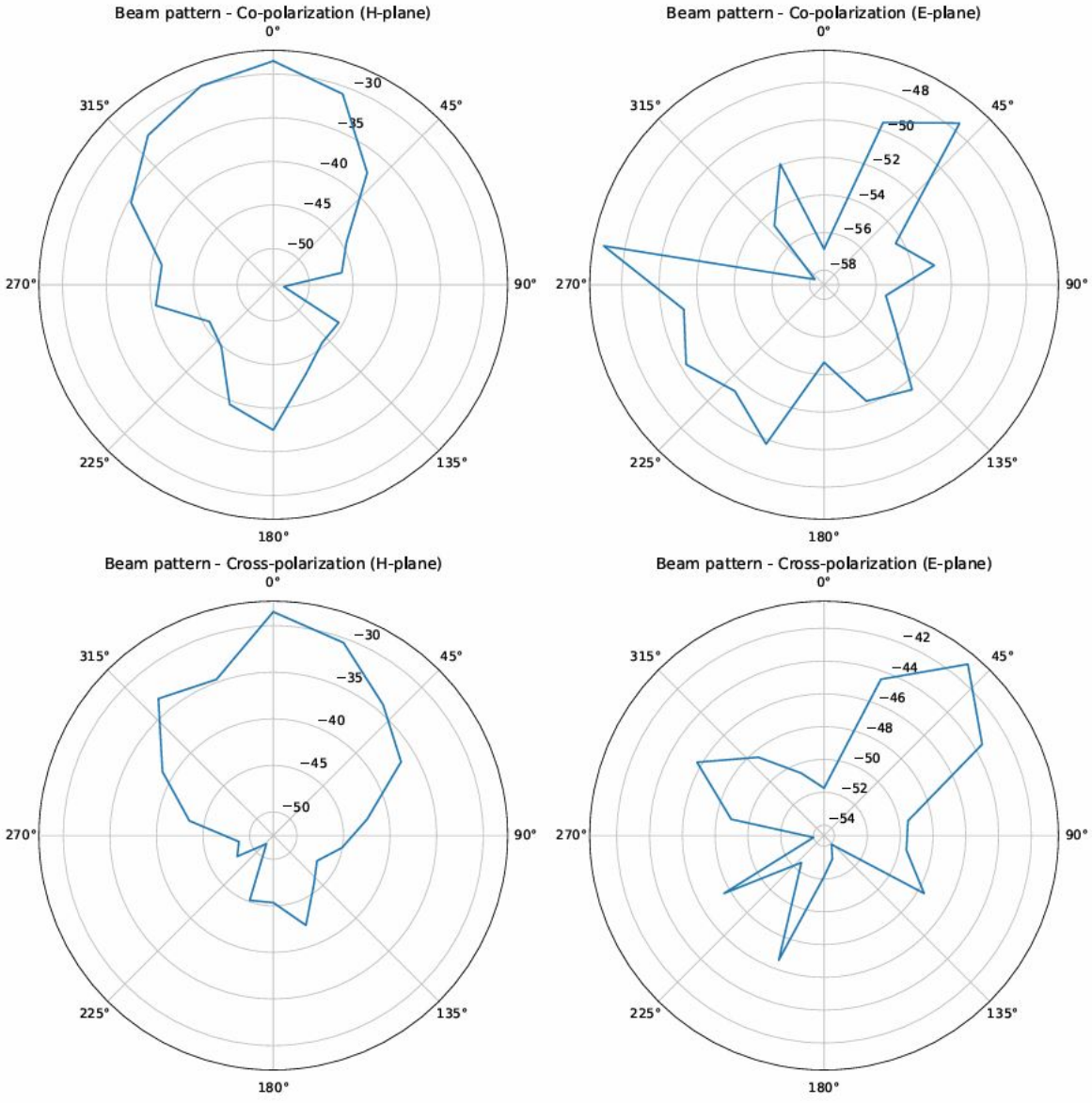


Figure 10: Antenna beam patterns in polar coordinates

Table 3: Angular received power measurements for antenna beam patterns

Angle $(^\circ)$	**Co-polarization (dBm)**		**Cross-polarization (dBm)**	
	E-plane	**H-plane**	**E-plane**	**H-plane**
0	-56.87	-28.50	-51.75	-28.52
20	-49.59	-30.88	-44.50	-30.50
40	-47.55	-37.39	-41.00	-34.20
60	-54.38	-44.46	-43.51	-36.70
80	-52.82	-46.17	-49.45	-42.30
100	-55.44	-52.92	-49.55	-45.05
120	-54.42	-45.48	-47.60	-47.12
140	-51.49	-45.45	-53.95	-45.75
160	-52.19	-43.24	-53.15	-42.30
180	-54.65	-37.49	-52.15	-45.35
200	-49.76	-39.55	-46.56	-45.15
220	-51.39	-44.88	-52.50	-51.40
240	-50.30	-45.70	-47.60	-48.07
260	-51.20	-40.47	-54.00	-48.80
280	-46.87	-41.17	-48.90	-43.40
300	-58.21	-35.32	-45.70	-38.80
320	-54.67	-31.80	-48.41	-33.35
340	-51.93	-29.90	-50.60	-34.70
360	-56.87	-28.50	-51.75	-28.52

1.7.5 Part E: Half-power beamwidth and directivity

1. From the co-polarized E-plane beam pattern, identify the two -3 dB points and calculate the E-plane HPBW θ_E.
2. From the co-polarized H-plane beam pattern, identify the two -3 dB points and calculate the H-plane HPBW θ_H.
3. Convert θ_E and θ_H to radians and estimate the directivity using

$$D \approx \frac{4\pi}{\theta_E \, \theta_H}. \tag{10}$$

The 3 dB threshold used to determine the angular width of the main lobes is visualized as a red dashed concentric line on the co-polarization beam patterns.

Linear interpolation of the co-polarization arrays yielded the following half-power beamwidths:

- **E-plane HPBW** (θ_E) :31.4° (0.548 rad). (Note: The main lobe in the E-plane exhibited a 45° mechanical offset relative to the true electrical peak during the experimental setup).
- **H-plane HPBW** (θ_H) :58.7° (1.024 rad).
- **Directivity** (D) : $\frac{4\pi}{\theta_E \theta_H} = \frac{4\pi}{0.548 \times 1.024} = 22.4$

Converting the absolute directivity ratio to decibels gives an estimated directivity of approximately **13.5 dBm**.

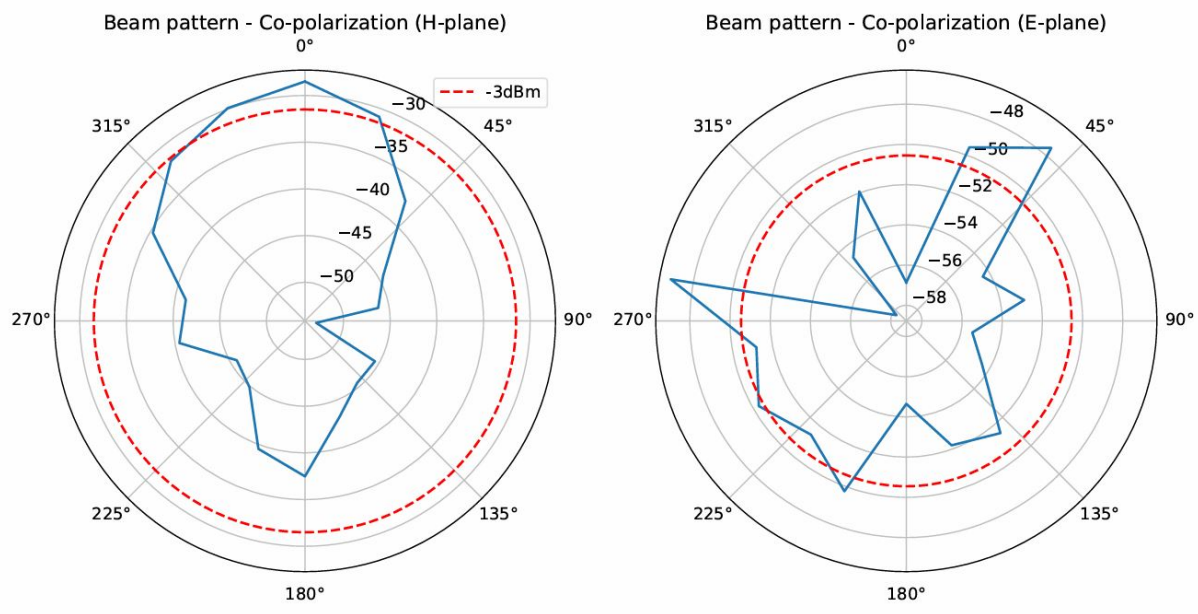


Figure 11: Antenna beam patterns in polar coordinates

1.8 Conclusions

The antenna characterization experiment successfully evaluated the radiation properties and path loss of the provided antenna. The key findings are summarized below:

- **Measured antenna operating frequency, minimum VSWR, and return loss (Part A)** : The theoretical operating frequency for the 5 cm quarter-wave monopole was calculated 1.5GHz. The verified far-field distance is 84.1 cm boundary , so all angular measurements are taken beyond this 84.1cm boundary. The minimum VSWR was found at frequency 220 MHz which is 1.08 . Minimum VSWR and maximum return loss indicate good impedance matching, but they do not guarantee that an antenna radiates efficiently. An antenna may absorb power in resistive losses instead of radiating it.

- **Effect of Distance (Part B)** : The received power decayed logarithmically with distance. Plotting the power against $20 \log_{10} R$ produced a highly linear negative slope, experimentally verifying that the antenna's far-field path loss strictly follows the inverse-square law.

- **Beam Patterns and Polarization (Parts C and D)**: The co-polarization measurements revealed distinct main radiation lobes in both the E-plane and H-plane. Conversely, the cross-polarization configuration resulted in significantly lower received power across all angles and lacked a defined main beam. This stark contrast demonstrates the antenna's strong polarization selectivity.

- **HPBW and Directivity (Part E)**: Using linear interpolation, the half-power beamwidths were determined to be θ_E =31.4° and θ_H =58.7°. This yielded an estimated directivity of $D \approx 22.4$, or 13.5 dBm, indicating a highly directional radiated beam.

- **Measurement Uncertainty**: The E-plane co-polarization plot exhibited a $40°$ main lobe shift, which was an artifact of mechanical misalignment during the physical setup. Additionally, the jagged secondary lobes observed in the polar plots are primarily attributed to multipath interference (RF reflections from the indoor laboratory environment) and the discrete $20°$ angular resolution of the measurements.